

The Art of Learning: Abstraction, Formalization, and the Climb from Word to World

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Abstract

Learning is commonly understood as the acquisition of knowledge. This paper proposes a different account: **learning is the act of climbing the abstraction mountain**—the structured ascent from phenomena to names, from names to techniques, from techniques to methods, from methods to formulas, from formulas to theories, from theories to formalizations, and from formalizations to understanding. We define **formalization** explicitly: the act of reconstructing received knowledge as a precise, stated structure—naming the object, stating the relations, defining the terms, proving the results, generalizing the structure, and finding the limits. We identify the **foundational layer** from which all learning begins: the senses, which provide direct contact with the world, and transmission, which provides knowledge from those who already have it—whether **overtly** (through explicit instruction, writing, formal teaching) or **covertly** (through imitation, observation, cultural absorption, and tacit apprenticeship). We identify **dogma** as received knowledge that has not yet been formalized—knowledge accepted on authority, defended by social sanction, and resistant to the very questioning that formalization requires. We map **kinds of knowledge** (know-what, know-how, know-why, and others) onto the levels, and we map **careers and apprenticeships** onto the climb. We advance and prove a hypothesis about human uniqueness: **humans enter the world at Level 1, while all other life enters at Level 0 or lower**. The proof is that language—the capacity for naming—is universal in *Homo sapiens*; universal, unlearned, species-typical capacities are biological endowments; therefore naming is a biological endowment of the species, and the human entry point to the mountain is Level 1. A central claim is that **the formalization of received knowledge is the engine of the climb**, and that dogma is the *arrest* of that engine. The history of civilization is the history of this pattern: each culture climbs higher because the previous culture formalized what it knew, and each culture plateaus when it dogmatizes what it received. The pattern is: **transmission** → **formalization** → **understanding** → **extension**. We then derive the **purpose of education** at four levels of aggregation (individual, society, nation, civilization) and map the **professions** (engineering, medicine, law, politics, business, vocational education, apprenticeship) onto the mountain. We observe that most education plateaus at Level 6 (formalization within a domain) and ask who goes beyond to Level 7 (understanding). We argue that the art of learning is the art of climbing the mountain—knowing which level you are on, which level the problem demands, how to travel between them, and how to formalize what you receive without dogmatizing it. We close by turning the framework on the coming century: AI is a **formalization engine**, and AI joined to robotics is the first artificial system that can complete the **round trip**—the loop between the world and the structure that is the engine of the climb itself. **Keywords:** learning, formalization, abstraction, dogma, education, professions, epistemology, philosophy of mind, artificial intelligence, robotics

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1 Introduction

1.1 The problem

What is learning? The standard answer: learning is the acquisition of knowledge. You learn facts, skills, methods. You accumulate. You get better. This answer is incomplete. It describes the *content* of learning without describing its *structure*. It says what is learned but not what learning *is*. This paper proposes a different answer. Learning is the act of **climbing the abstraction mountain**: moving from one level of abstraction to another, up or down. What is learned is the *level*, not the fact. The fact is an instance; the level is the structure.

1.2 The foundational layer

Before the mountain, there is a foundation. **All learning begins from two sources:**

1. **The senses.** Direct contact with the world. You see, hear, touch, taste, smell. The senses provide the raw material of experience from which all knowledge is constructed.
2. **Transmission.** Knowledge from those who already have it. This comes in two forms:
 - **Overt transmission:** explicit instruction, writing, teaching, formal communication. The knowledge is stated, and the receiver is told it is knowledge.
 - **Covert transmission:** imitation, observation, cultural absorption, tacit apprenticeship. The knowledge is demonstrated or embedded in practice, and the receiver acquires it without being explicitly told.

The senses are the foundational layer—they are how you first encounter the world. Transmission is the second layer—it is how you receive knowledge from others who have already climbed. Together, they are the **inputs** to the mountain.

1.3 Dogma: received knowledge that is not formalized

There is a third thing that enters the mountain. It is **dogma**.

Definition 1.1 (Dogma). *Dogma is received knowledge that has not been formalized, and that is defended against formalization.*

Dogma is not false knowledge. It may be true. It may be useful. What makes it dogmatic is not its truth value but its *epistemic status*: it is accepted on authority, defended by social sanction, and resistant to the questioning that formalization requires. Formally, dogma is the *arrest* of the engine. The engine says: received knowledge $\rightarrow$ formalization $\rightarrow$ understanding $\rightarrow$ extension. Dogma says: received knowledge $\rightarrow$ authority $\rightarrow$ enforcement $\rightarrow$ reproduction. Dogma is the failure mode of transmission. It is what happens when received knowledge stops being material for the climb and becomes the *substitute* for the climb.

1.4 Formalization: the central act

This paper uses “formalization” as its central term. It must be defined precisely.

Definition 1.2 (Formalization). *Formalization is the act of reconstructing received knowledge as a precise, stated structure.*

Formalization has six components:

1. **Naming the object.** What is this knowledge *about*? What is the structure?
2. **Stating the relations.** What are the relations between the components? How can they be expressed precisely?
3. **Defining the terms.** What are the primitives? What are the derived concepts?
4. **Proving the results.** Why do they hold? What are the justifications?
5. **Generalizing the structure.** What is the general form? To what other domains does it apply?
6. **Finding the limits.** Where does it break? What are the boundary conditions?

There is a fully rigorous way to state the same act. A **signature** σ fixes a vocabulary: sorts, and function and relation symbols with arities, from which terms and formulas are built. A **formal system** $(T, \vdash)$ is a set T of formulas—the axioms—together with the derivability relation $\vdash$ generated by the rules of inference, so that a proof is a finite sequence witnessing $T \vdash \varphi$. Against this backdrop the six components are precise operations:

- *Naming the object* is fixing the signature: the primitives of the knowledge become the symbols of the language.
- *Stating the relations* is writing the formulas: the claims of the knowledge become sentences of σ .
- *Defining the terms* is adding *conservative* definitions—eliminable extensions that license no new theorem in the old vocabulary.
- *Proving the results* is deriving the claims: $T \vdash \varphi$ for each intended result φ .
- *Generalizing the structure* is having the theorems hold in every model: $M \models \varphi$ for all $M \models T$, so the results state the structure, not an accident of one example.
- *Finding the limits* is discovering what the theory cannot decide: φ is independent of T when $T \not\vdash \varphi$ and $T \not\vdash \neg\varphi$. By Gödel's theorems, every consistent formalization strong enough to express arithmetic has such sentences and cannot prove its own consistency.

So formalization, precisely, is the construction of a formal system that receives knowledge, represents it as a signature and a set of sentences, proves its results, and is bounded by exactly the sentences it cannot derive. (A rigorous development, with the rectangle example expanded, is given in [Formalization, rigorously](#).) Formalization transforms received knowledge into understood knowledge. It converts the technique into a formula, the formula into a theory, the theory into a formalization. **Formalization is not the same as notation.** Notation is the *vehicle* of formalization—the symbols and syntax in which the structure is expressed. Formalization is the *act* of constructing the structure itself. **Formalization is not the same as mathematization.** Mathematization is formalization *in the language of mathematics*. Formalization is broader: it includes any precise, stated reconstruction of a structure. **Formalization is not the same as definition.** A definition specifies a single term. Formalization specifies an entire structure—its objects, relations, terms, results, generalizations, and limits.

1.5 An example of formalization

Consider the knowledge of *how to find the area of a rectangular field*. **Received knowledge (Level 2, technique)**. An Egyptian scribe knows: “To find the area, measure the length, measure the width, and multiply.” The scribe can do this. The scribe may not know *why* multiplication gives the area. The knowledge is procedural, transmitted by apprenticeship, opaque. **Formalization (Level 4, formula)**. A Greek geometer reconstructs the knowledge:

1. **Naming the object.** The object is a *rectangle*—a four-sided figure with four right angles.
2. **Stating the relations.** The area of a rectangle stands in a relation to its length and width. The relation is: $area = length \times width$.
3. **Defining the terms.** *Length* is the measure of one side. *Width* is the measure of the adjacent side. *Area* is the measure of the region enclosed.
4. **Proving the result.** Divide the rectangle into unit squares. The number of unit squares is the product of the number along the length and the number along the width. Therefore, $area = length \times width$.
5. **Generalizing the structure.** The same relation holds for any rectangle, in any units, at any scale. It generalizes to the area of any parallelogram (base $\times$ height), and to the area of any region approximated by rectangles.
6. **Finding the limits.** The relation fails for non-rectangular regions. It requires a theory of limits (integration) to extend to curved regions.

What has changed. Before formalization, the scribe had a *procedure*. After formalization, the geometer has a *structure*—an object, a relation, a proof, a generalization, and a boundary. The structure can be taught, extended, composed, and applied to new domains. The procedure can only be practiced. This is formalization: the transformation of a procedure into a structure.

1.6 The mountain

The mountain has levels. Each level is a stage of abstraction. Each level is a *separation of relations from substrate*.

Level	Name	What it is	Example
0	World	Phenomena, undefined	“That thing”
1	Word	Named phenomena	“Rock”
2	Technique	Ways of doing	“How to stack rocks”
3	Method	Systematic procedures	“Masonry”
4	Formula	Stated relations	“ $F = ma$ ”
5	Theory	Systems of formulas	“Newtonian mechanics”
6	Formalization	Structures of theories	“Symplectic manifold”
7	Understanding	Seeing the whole mountain	“Why mechanics works”

The levels are cumulative. Each level depends on the one below. Each level enables the one above.

1.7 The thesis

Learning is movement on the mountain. It is not the acquisition of facts; it is the *climb*—the ascent from phenomenon to understanding, and the descent from understanding to artifact. **Dogma is the arrest of movement.** It is the point at which received knowledge stops being transformed and starts being enforced. **Formalization is the engine of movement.** It is the act that transforms received knowledge into understood knowledge, and understood knowledge into extended knowledge.

1.8 The engine of the climb

The *mechanism* that drives the climb is: **The formalization of received knowledge is the engine of the climb.** Knowledge that is received but not formalized can only be practiced. Knowledge that is formalized can be understood, extended, and taught. The transition from received to formalized is the transition that enables the climb. It is the mechanism that turns a plateau into an ascent. It is why some cultures climb and others cycle. And dogma is the *blockage* of this mechanism. It is what happens when the transition is forbidden. The engine can be defined precisely, in the language of [Formalization, rigorously](#).

Definition 1.3 (The engine of the climb). *The engine of the climb is the map E that takes received knowledge K —a set of claims s , whether from the senses or from transmission—and returns a formalization $E(K) = (T, \vdash, \sigma, [\cdot])$ of it: a theory T in a signature σ , with a translation $[s]$ of each received claim, adequate for K in the four senses of [Formalization, rigorously](#)—inclusion, soundness, guard, and decisiveness.*

Four properties of the engine follow from the definition and make it a legitimate target for automation:

1. **It is a map.** The engine has a defined input (received knowledge K) and a defined output (an adequate formalization $E(K)$). Whatever else is creative about cognition, this step is a transformation—and a transformation can in principle be run by another system.
2. **Its correctness is checkable.** The output is adequate exactly when four formal tests pass: inclusion ($T \vdash [s]$ for every claim s), soundness (every theorem holds in every model), guard (definitions are conservative), and decisiveness (nothing essential is left independent). Verifiable conditions—checkable by computation once T , σ , and $[\cdot]$ are fixed.
3. **It reaches a limit.** Decisiveness is not always attainable: by Gödel’s first incompleteness theorem, every consistent formalization strong enough for arithmetic leaves independent sentences. A correct engine therefore reports not only what it derived but where it cannot decide—its independence boundary. Finding the limits is one of the six components, not a failure of the engine.
4. **Its output is not yet knowledge.** The engine produces a *candidate*: a structure that holds together and whose theorems are true in its models. Whether it holds *of the world* is decided by the descent—the experiment, the instantiation, the closed round trip. The engine drives the ascent; validation closes the descent.

This is the precise sense in which the engine is the engine: it is the transformation that turns received into understood; its correctness is formal and checkable; its output is sound within its models and bounded by its independence boundary; and it leaves the verdict on the world to the descent.

1.9 The human hypothesis

We advance and prove a hypothesis about the human species.

Hypothesis 1.4 (Human entry level). Humans enter the world at Level 1, while all other life enters at Level 0 or lower.

1.9.1 The proof

1. **Premise 1.** Every member of *Homo sapiens* has language—not as an achievement, but as a species-typical capacity. Every neurologically typical human child acquires language without instruction, universally, in the first years of life.
2. **Premise 2.** Language is the capacity to name—to refer to phenomena by symbols, and to create objects through those symbols. This is Level 1.
3. **Premise 3.** A capacity that is species-typical, universal, and acquired without instruction is a biological endowment, not a cultural achievement.
4. **Conclusion.** Therefore, *Homo sapiens* enters the world at Level 1. Naming is innate to the species, not learned by the individual.

1.9.2 The evidence for Premise 1

- **Every human society has language.** No human group has ever been found without one. This is true across every continent, every climate, every subsistence mode.
- **Every neurologically typical child acquires language.** Without formal instruction. Without conscious effort. On a predictable timetable.
- **Language acquisition has a critical period.** If a child is not exposed to language before puberty, the capacity does not develop normally. This is evidence of a *biological* window, not a cultural one.
- **Language has universal structural properties.** All languages have nouns and verbs, subjects and predicates, recursion, phonology, syntax. The surface differs; the deep structure is shared.
- **Language is not correlated with general intelligence.** Children with severe cognitive impairments can acquire language. Children with specific language impairment can have normal intelligence. Language is a *module*, not a general capacity.

1.9.3 The evidence for Premise 3

This is a general principle from biology and cognitive science:

- A capacity present in all members of a species, across all environments, without instruction, is *species-typical*.
- A species-typical capacity is *biological*—part of the organism’s innate endowment, not something the organism learns.
- Learning can refine and extend a biological capacity, but it cannot create it from nothing.

Examples: bipedalism in humans, echolocation in bats, web-spinning in spiders. All universal, unlearned, biological.

1.9.4 What the proof establishes

1. **Humans are born at Level 1.** The entry point to the mountain is not Level 0; it is Level 1.
2. **Non-human animals are born at Level 0 or below.** They have sensation but not naming. They do not climb the mountain—they live at Level 0.
3. **The human difference is the naming capacity.** Not intelligence, not consciousness, not tool use—naming.
4. **The possibility of the mountain is biological; the actual mountain is cultural.** The capacity to name is given; the capacity to formalize is not. The first step is automatic; every subsequent step is a climb.
5. **The difference between Level 0 and Level 1 is the difference between *being in the world* and *representing the world*.** Non-human life is in the world. Humans represent it.

1.9.5 What the proof does not establish

- It does not establish that humans are *superior*. It establishes that humans have a specific capacity—naming—that non-human life does not have.
- It does not establish that language is the *only* human endowment. There may be others.
- It does not establish that all humans reach higher levels. The entry is mandatory; the climb is optional.
- It does not establish the *mechanism* of the endowment. Whether language is a “module,” an “organ,” or an emergent property is an open question in cognitive science.

2 The foundational layer: senses, transmission, dogma

2.1 The senses

The senses are the foundational layer of all learning. They provide:

- **Raw material:** the phenomena that will be named, categorized, and formalized.
- **Direct contact:** unmediated access to the world, before any interpretation.
- **Feedback:** the ground truth against which all higher levels are tested.

The senses are the *empirical foundation*. Their limits: local, finite, noisy, private.

2.2 Transmission

2.2.1 Overt transmission

Overt transmission is **explicit**. The knowledge is stated, and the receiver is told it is knowledge. Examples: a teacher explaining a concept, a book stating a theorem, a manual describing a procedure. Powerful because it is **direct**. Limited because **it transmits the statement, not the understanding**.

2.2.2 Covert transmission

Covert transmission is **implicit**. The knowledge is demonstrated or embedded in practice. Examples: a child learning language by immersion, an apprentice learning a craft by watching a master, a newcomer absorbing a culture's norms. Powerful because it is **deep**. Limited because **it is opaque**. The old way of teaching was divided into two, and the two institutions correspond exactly to the two transmission modes. The *school* transmitted by the word—explicit, textual, addressed—and taught the pupil to *say*: to name, to recite, to state, without necessarily being able to do. The *workshop* transmitted by watching and doing—implicit, imitative, embodied—and taught the apprentice to *do*: the craft that passes from hand to hand, never written, never stated. The school taught what could be spoken; the workshop taught what could not. The two old ways of teaching are the two transmission modes made into institutions: the word and the work.

2.3 Dogma

Definition 2.1 (Dogma). *Dogma is received knowledge that has not been formalized, and that is defended against formalization.*

Dogma has three properties:

1. **Received.** It comes from transmission—overt or covert.
2. **Unformalized.** It has not been reconstructed from first principles.
3. **Defended.** It is protected from questioning. The defense may be social, institutional, or psychological.

Dogma is not the same as received knowledge. Received knowledge is the *material* for the climb. Dogma is the *arrest* of the climb. Dogma is not the same as false knowledge. A dogmatic belief may be true. Dogma is not the same as faith. Faith may be held with awareness that it is not formalized. Dogma claims it needs no formalization.

2.4 Why dogma blocks the climb

The engine of the climb is formalization. Formalization requires questioning: you must ask what the knowledge is, why it works, what it implies, where it breaks. Dogma forbids the questioning. The consequence:

- **Without formalization, there is no understanding.**
- **Without understanding, there is no extension.**
- **Without extension, the climb stops.**

Dogma is the arrest of the engine.

2.5 The three inputs

The mountain has three inputs:

- **Senses:** direct experience, first-person, empirical.
- **Transmission:** knowledge from others, second-person, explicit or implicit.
- **Dogma:** received knowledge defended against formalization.

The first two are *material for the climb*. The third is a *blockage of the climb*.

2.6 The pre-mountain state

Before the climb, you have senses, transmission, and possibly dogma. You have not yet started to climb. The climb begins when you start to **formalize what you receive**. The first step is naming: turning sensation into words, turning experience into concepts. **The human hypothesis** says that humans enter at Level 1: they already have the naming capacity, and the climb begins from there. All other life enters at Level 0 or lower: they have sensation but not naming, and they do not climb the mountain at all—they live at Level 0 in a state of direct, pre-linguistic cognition.

3 The levels, characterized

Each level has a *cost* and a *capability*. The cost is what it takes to reach the level. The capability is what the level enables.

3.1 Level 0: World

What it is: Phenomena. Undefined, contextual, analogical. **Input:** Senses. **Cost:** None. **Capability:** Direct experience. **Limitation:** No communication, no transmission, no accumulation. **Kind of knowledge:** Pre-conceptual. **Who is here:** All life.

3.2 Level 1: Word

What it is: Named phenomena. The name creates an object. **Input:** Senses + overt transmission. **Cost:** None for humans—the capacity is innate. **Capability:** Reference, discussion, transmission. **Limitation:** The name is undefined. It permits analogy, not precision. **Kind of knowledge:** Know-what. **Who is here:** Humans enter here.

3.3 Level 2: Technique

What it is: A way of doing. A procedure that works. **Input:** Senses + covert transmission. **Cost:** Practice. **Capability:** Effective action. **Limitation:** Opaque. Cannot be generalized or improved. **Kind of knowledge:** Know-how. **Who is here:** Apprentices.

3.4 Level 3: Method

What it is: A systematic procedure. **Input:** Senses + covert transmission + reflection. **Cost:** Reflection on technique. **Capability:** Transferability. **Limitation:** Procedural. Says *how*, not *why*. **Kind of knowledge:** Know-how, know-when, know-where. **Who is here:** Journeymen.

3.5 Level 4: Formula

What it is: A stated relation. **Input:** Overt transmission + abstraction. **Cost:** Abstraction from procedure. **Capability:** Precision, computation, generalization. **Limitation:** Isolated. Doesn't connect to other formulas. **Kind of knowledge:** Know-why. **Who is here:** Craftsmen transitioning to specialists.

3.6 Level 5: Theory

What it is: A system of formulas. **Input:** Overt transmission + synthesis. **Cost:** Synthesis. **Capability:** Explanation, prediction. **Limitation:** Framework-bound. **Kind of knowledge:** Know-why, know-what-if. **Who is here:** Specialists. Experts.

3.7 Level 6: Formalization

What it is: The structure of a theory, separated from its content. **Input:** Overt transmission + meta-abstraction. **Cost:** Meta-abstraction. **Capability:** Generality, composability, extendability. **Limitation:** Abstract. **Kind of knowledge:** Know-why-why. **Who is here:** Masters. Formalizers.

3.8 Level 7: Understanding

What it is: Seeing the whole mountain. **Input:** All sources, integrated, and dogma recognized. **Cost:** The entire climb. **Capability:** Movement between levels. Seeing limits. Seeing dogma for what it is. **Limitation:** Never complete. **Kind of knowledge:** All kinds, integrated, and the knowledge of limits. **Who is here:** Sages. Masters of the climb.

3.9 Technology: what the word points to

The ordinary word “technology” is not a level; it is a range. It names the lower levels of knowledge together with the tools that carry them:

- **Technique (Level 2).** The way of doing. Technology begins as a working procedure.
- **Method (Level 3).** The systematized procedure. A technology becomes transferable when the technique is turned into a method.
- **Formula (Level 4).** The stated relation. A technology gains precision, prediction, and control when the method is abstracted into a formula.
- **Tools.** The artifacts—instantiated at Level 0—in which technique, method, and formula are embedded. A hammer encodes a technique; a printed procedure encodes a method; a lever encodes a formula. Tools make execution easier, give control over the process, and make the outcome more certain.

So “technology” means: **the lower mountain—technique, method, and formula—made reusable through tools.** The ascent produces the knowledge; the descent (creation) produces the tools; the round trip is what we call technology.

4 Learning as movement

The key insight: **learning is not the acquisition of content; it is the movement between levels.**

- When you “learn” a fact, you are moving from Level 0 to Level 1. The fact is the name.
- When you “learn” a skill, you are moving from Level 1 to Level 2. The skill is the procedure.

- When you “learn” a method, you are moving from Level 2 to Level 3. The method is the systematized procedure.
- When you “learn” a formula, you are moving from Level 3 to Level 4. The formula is the stated relation.
- When you “learn” a theory, you are moving from Level 4 to Level 5. The theory is the system of formulas.
- When you “learn” a formalization, you are moving from Level 5 to Level 6. The formalization is the structure of the theory.
- When you “learn” understanding, you are moving to Level 7. Understanding is the ability to move.

Learning is not the content; it is the climb.

4.1 A rigorous definition

We can now define learning precisely, in the language of [Formalization, rigorously](#). First the state space. A *knowledge state* on the mountain is a level $\ell \in \{0, \dots, 7\}$; a climb is a sequence of states $(\ell_0, \ell_1, \dots, \ell_t)$ in which every step is one of two operations:

- **Ascent** $\ell_r \rightarrow \ell_{r+1}$ with $\ell_{r+1} > \ell_r$ —*formalization*, the engine. Received knowledge at ℓ_r is reconstructed as a theory at ℓ_{r+1} . In adequacy terms: each received claim s is translated to a sentence $[s]$ of the new signature, and $T \vdash [s]$ (inclusion); every theorem of T is true wherever its axioms are (soundness); and the axioms decide what the received knowledge claimed (decisiveness).
- **Descent** $\ell_r \rightarrow \ell_{r+1}$ with $\ell_{r+1} < \ell_r$ —*validation*, the confirmation. A claim formalized at ℓ_r is carried down to an intended model $M \models T$ and checked against Level 0 by experiment or instantiation.

Definition 4.1 (Learning). *Learning is a climb on the abstraction mountain: a sequence of knowledge states generated by ascent and descent, in which ascent is formalization and descent is validation, which begins from the senses and from transmission, and which completes a round trip—the claim formalized on the way up is validated on the way down.*

The definition makes the thesis precise. A fact is learned when its name is instantiated; a skill, when its procedure is executed; a theory, when its prediction is verified at Level 0. What is learned is the claim that completed the round trip. Movement that never descends produces candidates, not knowledge; movement that never ascends produces practice, not understanding. Both are failure modes of the climb.

5 Kinds of knowledge and the climb

Different kinds of knowledge live at different levels.

Kind of knowledge	What it is	Level(s)
Know-what	Declarative facts, categories, names	1
Know-how	Procedural skill, tacit ability	2–3
Know-when	Timing, context, appropriateness	3
Know-where	Location, domain, boundaries	3
Know-why	Explanation, causal understanding	4–5
Know-what-if	Counterfactual, predictive	5
Know-why-why	Structural, foundational	6
Know-who	Who knows what, who can help	1–7 (spanning)
Know-that-you-don't-know	Recognition of limits	7

Key insight: most knowledge is one kind at one level. The master at Level 7 has all kinds, integrated. Dogma fixes one kind at one level and forbids the others.

6 Careers and apprenticeships on the mountain

6.1 The generic career path

Career stage	Typical level	What it is	Failure mode
Novice	0–1	Learning names, categories	Never climbs past naming
Apprentice	2	Learning techniques by imitation	Never formalizes
Journeyman	3	Applying methods across contexts	Confuses method with formula
Craftsman	3–4	Refining methods, beginning to abstract	Stops at Level 4
Specialist	4–5	Mastering formulas and theories	Dogmatizes the domain
Expert	5–6	Synthesizing, formalizing	Never descends
Master	6–7	Seeing the whole, teaching	Dogmatizes the framework

6.2 The craft apprenticeship

Lives at Levels 2–3. The craft master may never climb above Level 3. This is not a failure—it is a *specialization*. The craft master is the *carrier* of tacit knowledge. The risk: the knowledge dies with them.

6.3 The scientific apprenticeship

Lives at Levels 4–7. The scientific master is the *carrier* of formalized knowledge. The risk: dogmatizing the framework.

6.4 The engineering career

Spans Levels 2–6. Engineering is the *descent*—taking formalized knowledge and instantiating it. The risk: applying formulas without understanding.

6.5 The managerial career

Spans Levels 1–6. Management is *coordination*. The risk: operating at Levels 2–3 without understanding.

6.6 The artistic career

Spans Levels 0–7. Art is *embodiment*. The risk: dogmatizing a school or style.

6.7 The scholarly career

Spans Levels 4–7. The scholar is the *carrier of formalized tradition*. The risk: dogmatizing the canon.

6.8 The pattern across careers

Every career is a path up the mountain. Every career has a typical level, and a typical ceiling. Every career has a failure mode—the level at which it stops climbing and starts defending. The art of learning, applied to a career, is:

- **Knowing the level you are at.**
- **Knowing the level your role requires.**
- **Knowing the level you could reach.**
- **Knowing the dogma that would stop you.**

And the master, in any career, is the one who can move between levels at will.

7 The two directions

7.1 Ascent: from phenomenon to understanding

The ascent is **problem-solving**. You encounter a phenomenon (Level 0), name it (Level 1), develop techniques (Level 2), systematize them (Level 3), abstract them into formulas (Level 4), synthesize formulas into theories (Level 5), formalize the theories (Level 6), and understand the whole (Level 7). The ascent makes the problem *legible*.

7.2 Descent: from understanding to artifact

The descent is **creation**. You start with understanding (Level 7), derive formulas (Level 4), turn formulas into methods (Level 3), turn methods into techniques (Level 2), and instantiate the techniques as artifacts (Level 0). The descent makes the solution *real*.

7.3 The round trip

Problem-solving and creation are the two halves of the same act. The ascent makes the problem legible; the descent makes the solution real. The round trip is what humans call *work*.

7.4 The validation requirement: why Levels 5 and 6 must descend

Level 5 (theory) and Level 6 (formalization) are where the claims are greatest. A theory explains and predicts; a formalization states the structure of what is known. But reach is not truth. A claim at Level 5 or Level 6 is validated only by descending to Level 0—by experiment, observation, or instantiation.

- **Why prediction must descend.** A theory earns its place by predicting what the world does. A prediction is a claim about Level 0, and the only judge of a claim about the world is the world. The experiment is the down-climb that carries the claim to the world; the observation is the verdict. Newton’s theory descended to falling bodies and planetary orbits; relativity descended to the eclipse.
- **Why formalization must descend.** A formalization separates structure from content. An internally consistent formalization can still refer to nothing—its terms uninterpreted, its results true of no phenomenon. The descent regrounds the structure: the artifact, the measurement, the test either bears the formalization out or breaks it.
- **Why consistency is not enough.** Proof shows that a structure holds together; it does not show that the structure holds *of* anything. The round trip supplies the missing half.

So the ascent alone does not produce knowledge; it produces *candidates*. Knowledge is the claim that has survived the descent. This is the deepest form of never descending (failure mode two): the theory that is never tested, the formalization that is never instantiated.

8 The failure modes

8.1 Failure mode 1: Never climbing

You stay at Level 0 or 1. You point, name, and describe. You reason by analogy. You can do, but you cannot state. You plateau at the limit of your personal ability. **Examples:** Egypt. Babylon. Every practitioner who never formalizes.

8.2 Failure mode 2: Never descending

You climb to Level 6 or 7 and stay there. You formalize, but you never instantiate. You produce theory without application. **Examples:** Pure theorists who never build.

8.3 Failure mode 3: Climbing to the wrong level

You climb to the wrong level for the problem. You over-abstract or under-abstract. **Examples:** Most practical failures.

8.4 Failure mode 4: Dogmatizing

You receive knowledge and defend it against formalization. You stop climbing. You reproduce instead of extending. **Examples:** Every orthodoxy that chose reproduction over understanding. Every craft that died with its master. Every company that defended its model instead of formalizing it.

9 The historical pattern

9.1 Egypt and Babylon: technique

Egypt reached Level 2 (technique) and stayed. It had extraordinary practical mathematics—a working geometry of areas and volumes, bound to the annual surveying of the fields after the Nile’s flood. But it never abstracted to formula (Level 4) or theory (Level 5). Its mathematics was procedural, transmitted by apprenticeship, bound to specific problems. Babylon reached Level 2 and stayed, and is the earlier and more complete case. Babylon invented the base-60 numeral system with place value, arithmetic tables for multiplication and reciprocals, and procedures for solving problems we now recognize as quadratic equations. The Babylonians tabulated Pythagorean triples more than a millennium before Pythagoras. And yet the procedures were procedures: opaque, transmitted by apprenticeship, never reconstructed as stated structures with proofs, never generalized beyond the working problems on the clay tablets. Egypt plateaued for 3,000 years because it never climbed. Babylon plateaued for 2,000 years because it never climbed.

9.2 Greece: theory

Greece received Egyptian and Babylonian mathematics and *could not take it on authority*. The culture of public debate demanded justification. So the Greeks invented proof: the axiomatic method. The Greeks climbed from Level 2 (technique) to Level 5 (theory). They formalized the received results—the Egyptian geometry of areas, the Babylonian arithmetic and algebra—and exceeded them. Euclid’s *Elements* is the canonical text. Greece climbed because it had to.

9.3 The Islamic world: algebra

The Islamic world received Greek mathematics and Indian numerals, and with them the accumulated inheritance of Egypt and Babylon. It formalized all of it. Al-Khwarizmi’s algebra is the synthesis. The Islamic world climbed from Level 5 (theory) to Level 5+ (algebra, extension).

9.4 Europe: formalization

Europe received Greek and Islamic mathematics. It formalized both, and went further. Descartes, Fermat, Newton, Leibniz invented calculus—the formalization of change. Europe climbed from Level 5 (theory) to Level 6 (formalization).

9.5 Modern mathematics: understanding

Modern mathematics has climbed to Level 7 (understanding). It studies the structure of formalization itself: category theory, type theory, computation theory, model theory.

9.6 The pattern

Transmission → formalization → understanding → extension. Each culture climbs because it receives. The reception creates the need for formalization. The formalization produces understanding. The understanding enables extension. The mountain is *cumulative*. Each culture stands on the previous one’s shoulders. Each climb enables the next.

9.7 The failure mode

Transmission $\rightarrow$ **dogma** $\rightarrow$ **stagnation**. Each culture that dogmatizes what it receives stops climbing. The knowledge is preserved but not extended. The mountain becomes a plateau.

9.8 The scientific revolution: when the descent became inescapable

Before the scientific revolution, the widespread ways of learning throughout the world skipped formalization all together: they jumped from theory to “understanding.” A theory was received, and the reception itself was taken to be understanding—the student assented to the statement without reconstructing its structure. That understanding was then passed on through dogma: transmitted, defended against formalization, never tested. The scientific revolution broke the pattern twice. It took the old Greek approach—the axiomatic, proof-based method—and made formalization an inescapable step of the climb: a thing was not understood until it was stated as a structure with proofs. And it refused to hedge itself atop understanding using dogma: under the new method a theory could not stand simply because it was asserted and believed. The experiment—the climb-down that carries a claim back to Level 0 to be confirmed or falsified by the world—became another inescapable part of learning and of the generation of proven knowledge: science. In the scientific sense, knowledge is exactly the claim that has made the full round trip: formalized on the ascent, validated on the descent, and held, in between, without dogma.

10 The art of learning

We can now state the art of learning precisely. **The art of learning is the art of climbing the mountain.** It has nine components.

10.1 Knowing the sources

You must know that all learning begins from the senses and from transmission (overt and covert).

10.2 Recognizing dogma

You must know how to recognize dogma—received knowledge defended against formalization. The signs:

- **Social sanction:** questioning is punished or discouraged.
- **Institutional closure:** the knowledge cannot be examined by outsiders.
- **Psychological defense:** doubt is treated as betrayal.
- **Absence of justification:** the knowledge is asserted but not proven.
- **Resistance to formalization:** attempts to reconstruct it are rejected.

Recognizing dogma is the first step to liberating the knowledge it contains.

10.3 Knowing the kinds of knowledge

You must know the kinds of knowledge—know-what, know-how, know-when, know-where, know-why, know-what-if, know-why-why, know-who, and the knowledge of limits—and where each lives on the mountain.

10.4 Knowing the levels

You must know that the levels exist, what each enables, and what each costs.

10.5 Knowing where you are

You must know which level you are on—and which level your career, your role, and your organization are on.

10.6 Knowing what the problem demands

You must know which level the problem requires.

10.7 Knowing how to travel

You must know how to move between levels—both ascent and descent.

10.8 Knowing how to formalize what you receive

You must know how to take received knowledge and formalize it—naming, stating relations, defining terms, proving results, generalizing, finding limits.

10.9 Knowing not to dogmatize

You must know how to formalize *without dogmatizing the result*. The climb is not from dogma to formalization; it is from *any* fixed position to the next. Understanding is asymptotic.

11 The purpose of education

11.1 The structural prediction

If learning is the act of climbing the abstraction mountain, and if the engine of the climb is the formalization of received knowledge, and if dogma is the arrest of the engine, then **the purpose of education is to manage the climb**. Education is the institution that:

- **Produces climbers**—individuals who can move up the mountain.
- **Sustains the climb**—societies that formalize their received knowledge.
- **Compounds the climb**—civilizations that build on the past across generations.

11.2 The levels of education

The institutional ladder maps onto the mountain.

Education stage	Level	What it teaches
Pre-school	1	Names, words, early know-what
Primary / basic	1–2	Literacy, numeracy, and the first techniques—know-what, know-how
Secondary	2–3	Techniques and methods—know-how, know-when, know-where
Tertiary (undergraduate)	3–5	Formulas and theories—know-why
Advanced (master’s, doctorate)	5–6	The structure of a field; the doctorate certifies formalization—know-why-why
Beyond (research careers)	6–7	Movement between levels and the recognition of limits

Primary names the world; secondary systematizes the doing; tertiary systematizes the understanding; the doctorate certifies the formalization. The ladder therefore climbs exactly as far as Level 6 and then stops—no credential exists for Level 7. Education feeds the career ladder in step: primary produces novices, secondary produces apprentices, tertiary produces specialists, and advanced education produces masters.

11.3 The purpose for the individual

The purpose: to reach Level 7—understanding. **The means:** teaching the climb; teaching formalization; teaching dogma recognition; teaching the kinds of knowledge; teaching the levels. **The failure mode:** producing people who can name but not formalize. **The success mode:** producing people who can climb—who can move between levels, formalize what they receive, and recognize dogma.

11.4 The purpose for a society

The purpose: to reproduce and extend the society’s knowledge. **The means:** overt transmission; covert transmission; formalization. **The failure mode:** dogmatizing the tradition. **The success mode:** holding both reproduction and extension.

11.5 The purpose for a nation

The purpose: to produce a critical mass of climbers. **The means:** strategic education at all levels; institutions that support formalization; a culture that values formalization. **The failure mode:** import-dependence. **The success mode:** producing a critical mass of climbers who can sustain the climb and teach the next generation.

11.6 The purpose for a civilization

The purpose: to compound knowledge across generations. **The means:** institutions of formalization; the meta-level; continuity across generations. **The failure mode:** the plateau. **The success mode:** compounding across centuries.

11.7 The unifying claim

The purpose of education, at every level, is to manage the climb. It is the *infrastructure* of the climb.

11.8 The testable prediction

Prediction 11.1. *The trajectory of a society, nation, or civilization is determined by its education system's ability to produce climbers.*

- A society that formalizes its received knowledge climbs.
- A society that dogmatizes its received knowledge plateaus.
- A society that imports knowledge without formalizing it remains dependent.
- A society that produces a critical mass of climbers becomes a leader.

12 The professions

12.1 The general principle

A **profession** is a domain with a body of received knowledge, a practice, a formalization, an institution, and a failure mode. The position of a profession on the mountain is determined by: where it enters; where it peaks; what it formalizes; what it dogmatizes.

12.2 Engineering

Entry: 2–3. **Peak:** 6. **Formalizes:** the relation between form and function. **Failure mode:** applying formulas without understanding. **The descent:** engineering is the canonical descent—taking formalized knowledge and instantiating it as artifacts.

12.3 Medicine

Entry: 2. **Peak:** 6. **Formalizes:** the relation between mechanism and outcome. **Failure mode:** protocol without understanding. **The special challenge:** medicine is the most vertical profession—it spans Levels 0–6 in a single act.

12.4 Law

Entry: 1. **Peak:** 5. **Formalizes:** the relation between rule and case. **Failure mode:** formalism without justice. **The special challenge:** law is the most dogmatic-prone profession.

12.5 Politics

Entry: 1. **Peak:** 5–6. **Formalizes:** the relation between interest and outcome. **Failure mode:** ideology as dogma. **The special challenge:** politics is the most analogical profession.

12.6 Business

Entry: 1. **Peak:** 6. **Formalizes:** the relation between model and outcome. **Failure mode:** model without formalization. **The special challenge:** business is the most competitive profession.

12.7 Vocational education

Entry: 1. **Peak:** 3–4. **Formalizes:** the relation between technique and material. **Failure mode:** tacit knowledge that dies with the master.

12.8 Apprenticeship

Entry: 2. **Peak:** 3–4. **Formalizes:** the relation between practice and theory. **Failure mode:** the perpetual apprentice.

12.9 The synthesis

Profession	Entry	Peak	Formalizes	Failure mode
Engineering	2–3	6	Form–function	Formula without understanding
Medicine	2	6	Mechanism–outcome	Protocol without understanding
Law	1	5	Rule–case	Formalism without justice
Politics	1	5–6	Interest–outcome	Ideology as dogma
Business	1	6	Model–outcome	Model without formalization
Vocational	1	3–4	Technique–material	Tacit knowledge dies
Apprenticeship	2	3–4	Practice–theory	Perpetual apprentice

The pattern: professions that formalize climb; professions that dogmatize plateau. The formalization is the engine; the dogma is the arrest.

13 Beyond Level 6

13.1 The observation

Most education plateaus at Level 6. The university teaches formalization and then stops. The PhD certifies competence at Level 6. There is no institutional ladder beyond it. The question: who goes beyond Level 6 to Level 7?

13.2 What Level 7 is

Level 7 is *understanding*—seeing the whole mountain. It is the meta-level. It is the ability to:

- See the levels.
- Move between levels.
- See the limits.
- See dogma.
- Integrate the kinds of knowledge.
- Cross domains.
- Create new formalizations.

Level 7 is not a credential. It is a state. And it is rare.

13.3 Why Level 7 borders on philosophy

Level 7 is the highest level we can name, and it is not the end of the mountain. We do not know how high the mountain extends after Level 7—whether understanding above understanding exists, whether the meta-level itself has a meta-level, whether there is a structure to the whole that no position on it can see. This question cannot be settled by descent. It is not a claim about Level 0; no experiment can test the height of the mountain from a position on it. The questions that remain at the top are questions about the mountain itself—its top, its limits, its relation to what lies beyond it. That is why Level 7 borders on philosophy. At the top, the climb meets the limits of what can be known from inside the climb, and the remaining questions are not empirical but structural: the shape of knowledge, the possibility of further levels, the limits of formalization itself. The sage at Level 7 does not answer them. He sees that they are the questions, and that the mountain continues above the view.

13.4 Why complete knowledge of everything is impossible—and why knowledge of something is not

Complete and perfect knowledge of *everything* is impossible. This is not a doubt; it follows from the framework and from two limit theorems of logic.

1. **The whole is not in the view.** Complete knowledge of everything is knowledge of the whole mountain: every level, every relation, everything formalized, and whatever extends above Level 7. By the principle already established here—we do not know how high the mountain extends after Level 7, and Level 7 is defined by its limitation, never complete—no position on the mountain sees the whole.
2. **Reflexivity.** “Everything” includes the knowledge itself. A complete body of knowledge must know that it is complete—a claim about itself—and must certify its own truth. The claim “this is all of it, and it is all true” is precisely a claim about the whole made from inside the whole.
3. **Gödel.** No consistent formalization strong enough to express arithmetic decides every sentence of its language (first incompleteness theorem), and none can prove its own consistency (second). To claim one has formalized everything is to claim a decisive, self-certifying formalization reaching at least arithmetic—which the theorems exclude.
4. **Tarski.** No language adequate for arithmetic can define its own truth. The sentence “all of this is true” cannot be expressed in the theory it describes, let alone proved. An all-knowing system would have to say it.

The conclusion. Complete and perfect knowledge of everything is impossible in two independent ways: the whole is not visible from any position on the climb, and a complete, self-certifying formalization cannot exist. **But knowledge of something is not impossible.** A bounded structure, inside a fixed formal system, is perfectly knowable. The rectangle theorem is derived, decisive, and certain within its theory, and the theory knows its own theorems and its own boundaries. That is what formalization achieves at Level 6—perfect knowledge of a structure within its limits—and what understanding achieves at Level 7 locally: mastery of a bounded whole, never mastery of the totality. The epistemology of the mountain is therefore: knowledge is local, bounded, and perfectible; totality is impossible; and the only knowledge that is safe is the knowledge that knows its own bounds. Every claim to know everything is dogma—including the claim to have finished the climb.

13.5 The five types

1. The polymath. Crosses domains. Sees the structure of formalization itself. **2. The founder.** Descends and creates. Formalizes a domain and instantiates the formalization. **3. The teacher.** Transmits the whole. Teaches the climb, not just the content. **4. The sage.** Sees the limits. Knows what is formalizable and what isn't. **5. The child.** Enters fresh. Climbs naturally, without dogma. The renewal of the climb. These are not exclusive categories. A single person can be several. But for all five, the reach is the same: those who attain Level 7 are exceptional humans who define a humanity's era.

13.6 Why education plateaus at Level 6

Five structural reasons:

1. **The credential system ends at Level 6.** The PhD is the highest degree.
2. **The disciplines are siloed.** No department is responsible for the structure of formalization itself.
3. **The reward system rewards depth.** Academic careers are built on publications within a domain.
4. **The dogma of specialization.** "Know more and more about less and less."
5. **The absence of the meta-level.** No institution teaches the mountain.

13.7 The historical pattern

Every civilization that climbed produced Level 7 individuals—exceptional humans who define a humanity's era. But the production was exceptional, not systemic.

- Greece produced its exemplars—through culture, not system.
- The Islamic world produced its exemplars—through culture, not system.
- Europe produced its exemplars—through culture, not system.

The pattern: the system produces Level 6; the culture produces Level 7.

13.8 The prediction

Prediction 13.1. *A civilization that produces Level 7 individuals climbs higher than one that does not.*

The evidence: every civilization that led produced Level 7 individuals. Every civilization that plateaued did not. **The implication:** education should not stop at Level 6. It should produce Level 7—not as a credential, but as a state.

14 Implications

14.1 For individuals

Learning is not accumulation. It is movement. You learn by climbing the mountain, using your senses and the transmission from others as inputs, formalizing what you receive, and resisting the temptation to dogmatize it.

14.2 For organizations

Organizations have a level. The ceiling of an organization is the level it has reached—or the dogma it has adopted. Organizations grow by climbing. They plateau by stopping. They die by dogmatizing.

14.3 For civilizations

Civilizations climb over centuries. The pattern is: transmission → formalization → understanding → extension. The failure mode is: transmission → dogma → stagnation.

14.4 For careers

Every career is a path on the mountain. Knowing where your career is on the mountain—and where it could be—is the difference between a career that climbs and a career that plateaus.

14.5 For the future

The mountain has been climbed for millennia and it is still growing. The climb will not stop on its own—it stops only when received knowledge hardens into dogma. Knowing the art of learning, in every generation, is what keeps the climb alive.

15 The engine automated: AI and the mountain

The framework does not merely describe human learning. It predicts where artificial intelligence will land. The answer is not “everywhere.” It is more specific, and more interesting.

15.1 The short answer

AI has the most impact at Level 6 and the transition from Level 6 to Level 7. It is already transforming Level 4 and Level 5. It is beginning to touch Level 7. It is not touching Levels 0–3 in the same way, because those levels are grounded in the senses and in embodied practice, where AI has no direct access. But the deepest impact is at Level 6—formalization—because that is the level where AI is native. AI is a formalization engine. It operates on structure. And Level 6 is the level of structure.

15.2 Level 0: World—minimal impact

AI has no direct access to the world. It has access to data *about* the world—text, images, sensor readings—but not to the world itself. It cannot touch, taste, smell, or feel. It cannot be present. *What AI can do at Level 0:* process sensor data, control robots, interpret images. This is real, but it is mediated—AI operates on representations of the world, not the world. *What AI cannot do:* be in the world. Have direct experience. Ground its knowledge in sensation. **Impact:** real but indirect. AI extends human reach at Level 0 (through robotics, sensing, automation), but it does not enter Level 0.

15.3 Level 1: Word—significant but shallow impact

AI is very good at Level 1. It manipulates words, generates text, translates languages, classifies documents. Large language models are, at their core, Level 1 engines—they operate on symbols. *What AI can do at Level 1:* generate text, translate, summarize, classify, answer questions, produce

names and categories. *What AI cannot do:* understand what the words mean in the grounded sense. It operates on statistical relations between symbols, not on the referents. **Impact:** enormous in scale, limited in depth. AI has democratized Level 1—anyone can now produce fluent text. But it has not changed what Level 1 is.

15.4 Level 2: Technique—moderate impact

AI can learn procedures from data. It can perform tasks that were previously tacit—recognizing faces, driving cars, playing games, diagnosing images. *What AI can do at Level 2:* perform techniques, often better than humans, at scale, without fatigue. *What AI cannot do:* acquire techniques the way humans do—through embodied practice, imitation, and apprenticeship. AI learns from data, not from doing. **Impact:** substantial. AI is automating many Level 2 techniques. But it does so by *bypassing* the human path to Level 2, not by replicating it. **The key insight:** AI does not climb from Level 0 to Level 1 to Level 2 the way humans do. It enters at Level 2 (or Level 4) directly, trained on data. This is a fundamental difference.

15.5 Level 3: Method—significant impact

AI can systematize procedures. It can find patterns across cases, optimize workflows, generate methods. *What AI can do at Level 3:* design methods, optimize processes, discover heuristics from data. *What AI cannot do:* reflect on technique the way humans do. It does not notice the pattern in the doing; it extracts the pattern from the data. **Impact:** substantial. AI is becoming a method-designer—it can propose new procedures, new workflows, new optimizations. But it does so without the human’s reflective relationship to the practice.

15.6 Level 4: Formula—major impact

AI is a formula engine. It can derive relations, fit models, discover equations, generate proofs. *What AI can do at Level 4:* discover formulas from data (symbolic regression), prove theorems, generate equations, fit models, predict outcomes. *What AI cannot do:* understand why the formula holds. It can produce the formula; it cannot grasp its justification in the way a human geometer grasps the proof of the area of a rectangle. **Impact:** major. AI is transforming Level 4 across every domain—physics, biology, economics, engineering. It is becoming a formula-discovery engine at a scale no human could match. **The key insight:** AI’s strength grows with the level of structure. It operates on structure, and formulas are structure.

15.7 Level 5: Theory—major impact

AI can synthesize formulas into theories. It can find connections across domains, generate hypotheses, build models. *What AI can do at Level 5:* synthesize theories, connect formulas, generate hypotheses, propose frameworks. *What AI cannot do:* see the theory as a whole in the way a human master does. It can construct the theory; it cannot inhabit it. **Impact:** major. AI is beginning to do Level 5 work—generating hypotheses, synthesizing across domains, proposing new frameworks. This is the frontier of current AI capability.

15.8 Level 6: Formalization—the deepest impact

AI operates at every symbolic level of the mountain—word, formula, theory, understanding. But “native” should not be read as exclusive participation; it means *greatest comparative advantage*.

Every level that AI touches it touches well, but at Level 6—formalization—it produces, verifies, and recombines structure at a scale, speed, and breadth no human could match. That is the level where AI is native. Level 6 is the level of structure—the reconstruction of knowledge as a precise, stated structure. AI is a formalization engine. It operates on structure. It produces structure. It manipulates structure. *What AI can do at Level 6:* formalize knowledge at scale, reconstruct structures, generate formalizations, translate between formalisms, verify proofs, compose structures. *What AI cannot do:* understand the formalization in the human sense. It can produce the structure; it cannot grasp why the structure has the form it has. **Impact:** deepest. AI is transforming Level 6 across every domain. It can:

- Formalize knowledge at a scale no human could match.
- Reconstruct structures from data.
- Generate formalizations from examples.
- Verify proofs, check consistency, find errors.
- Translate between formalisms (e.g., from prose to mathematics).
- Compose structures (e.g., combine theories into larger theories).

This is where AI changes the mountain most. Not because it does what humans do, but because it does what humans do at Level 6—formalization—at a scale, speed, and breadth no human could match. **The consequence:** the climb from Level 5 to Level 6 is being automated. What took humans centuries (the formalization of physics, of biology, of economics) can now be done by AI in years, months, or days.

15.9 Level 7: Understanding—emerging impact

Level 7 is seeing the whole mountain. It is the meta-level. It requires:

- Seeing the levels.
- Moving between levels.
- Seeing the limits.
- Seeing dogma.
- Integrating the kinds of knowledge.
- Crossing domains.
- Creating new formalizations.

What AI can do at Level 7: cross domains (it already does), integrate knowledge (it does), generate new formalizations (it is beginning to). It can see connections across domains that humans miss. *What AI cannot do:* see the limits in the human sense. It cannot recognize dogma—including its own. It cannot have the knowledge of limits, because it does not have a grounded relationship to what it knows. It does not know what it doesn't know. **Impact:** emerging, and potentially enormous. AI is beginning to do Level 7 work—cross-domain synthesis, meta-level reasoning, the generation of new formalizations. But it does so without the grounding, the limits, and the self-awareness that characterize human Level 7.

15.10 The synthesis: where AI has the most impact

Level	AI impact	Why
0 World	Minimal	No direct access to the world
1 Word	Significant but shallow	Operates on symbols, but no grounding
2 Technique	Moderate	Can perform, but doesn't acquire like humans
3 Method	Significant	Can systematize, but doesn't reflect
4 Formula	Major	Operates on structure
5 Theory	Major	Operates on synthesis
6 Formalization	Deepest	Native to formalization
7 Understanding	Emerging	Can cross domains, but lacks grounding and limits

The prediction: AI's deepest impact is at Level 6—formalization—because that is the level where AI is native. It is a formalization engine. And Level 6 is the level that drives the climb. **The consequence for the mountain:** if AI automates Level 6, it automates the engine of the climb. The climb itself accelerates. Domains that took centuries to formalize can be formalized in years. The mountain grows faster.

15.11 The deeper implication

The framework predicts something specific about AI's role: **AI is not a climber. It is an engine.** Humans climb the mountain. They start at Level 1 (innate), and they climb through Levels 2–7 by formalizing what they receive. The climb is grounded in the senses, in transmission, in embodied practice, in the recognition of limits. AI does not climb. It does not start at Level 1. It does not have senses. It does not have transmission in the human sense. It does not have limits in the human sense. It does not have dogma in the human sense. What AI does is formalize. It takes received knowledge—in the form of data, text, examples—and reconstructs it as structure. This is exactly the engine of the climb. And AI does it at scale. So **AI is not a replacement for the climber. It is a replacement for the engine.** This has consequences:

1. **The climb accelerates.** If the engine is automated, the climb goes faster.
2. **The human role shifts.** Humans still climb—they still start at Level 1, they still ground in the senses, they still recognize limits. But they no longer have to do the formalization by hand. They can direct the engine.
3. **Level 7 becomes more important, not less.** If AI does Level 6, then the human's unique contribution is Level 7—understanding, limits, dogma recognition, the meta-level. The human becomes the director of the climb, not the engine of it.
4. **The dogma problem intensifies.** If AI formalizes everything, the temptation is to accept AI's formalizations as dogma. This is the new dogma. The human's role is to question it—to recognize when the AI's formalization is defended against questioning, and to formalize it anyway.
5. **The education system must adapt.** If AI does Level 6, then education should not stop at Level 6. It should produce Level 7—the understanding, the limits, the dogma recognition. The purpose of education shifts from producing formalizers to producing directors.

15.12 The prediction

Prediction 15.1. *AI's deepest impact will be at Level 6—formalization—because AI is a formalization engine. It will automate the engine of the climb. And this will shift the human role from doing formalization to directing it, and from Level 6 to Level 7.*

The consequence: the mountain grows faster, but the human contribution shifts upward. The human becomes the climber who sees the whole mountain, and the AI becomes the engine that formalizes along the way. **The risk:** if humans mistake AI's formalizations for dogma—if they accept the structures as given, without questioning—then the engine becomes the arrest. The formalization becomes dogma. The climb stops. **The opportunity:** if humans use AI to accelerate formalization while retaining the Level 7 capacity to question, to see limits, and to recognize dogma, then the climb accelerates without the arrest. The mountain grows, and the climbers remain in control.

15.13 Precisely: what the rigor predicts

The definition of the engine as a map $E: K \mapsto (T, \vdash, \sigma, [\cdot])$ turns these claims into derived predictions.

1. **The ascent is automatable in principle.** If the received knowledge K is codifiable—a computable set of claims—and the adequacy tests are checkable, then an automated engine $\hat{E}$ can in principle be implemented: search for a signature, axioms, and translation that pass inclusion, soundness, guard, and decisiveness. No step in the definition is one a machine cannot in principle run. The prediction: every domain whose received knowledge is codified becomes formalizable by machine, domain by domain.
2. **The bound is Gödel.** Decisiveness fails for every consistent formalization strong enough for arithmetic. A correct automated engine must therefore report two outputs: the theorems it derived *and* the claims it cannot decide. Accepting machine formalization while ignoring the declared boundary is not reliance on the engine; it is dogma by machinery. **Observable:** a well-built engine publishes its independence boundary with every output.
3. **The codifiability bound.** Knowledge that has never been stated—the covert, embodied, apprenticed technique—cannot enter the engine until it is turned into claims. The sensors and actuators that turn doing into data are part of the ascent: robotics, not only AI, makes the tacit codifiable. **Observable:** the domains an engine can climb are exactly the domains for which a codification pipeline exists.
4. **The descent is not formal.** Inclusion, soundness, guard, and decisiveness are conditions on the theory; none of them is a test against Level 0. Whether the candidate holds *of the world* is answered only by the experiment—the instance, the measurement, the closed round trip. Therefore **pure AI produces candidates, pure robotics produces practice, and only the pair produces proven knowledge.** This is the formal reason robotics is the crucial pairing.
5. **The rate of the climb accelerates.** The engine is a map with a checkable output: it is repeatable and parallelizable. Once automated, the rate of formalization is bounded by the search over structures, not by human labor. **Observable:** domains that took centuries to formalize are formalized in years; the constraint shifts from doing formalization to verifying it and setting its direction.

These five are the predictions the rigorous engine supports, and they are the predictions the robotics chapter uses.

15.14 The one-line statement

AI has the most impact at Level 6—formalization—because AI is a formalization engine, and Level 6 is the level of structure. It automates the engine of the climb. The human role shifts from formalizing to directing—from Level 6 to Level 7. And the danger is that AI’s formalizations become the new dogma, defended against the very questioning that would drive the next climb. This is the prediction, and it has practical consequences for education, for professions, and for the future of the mountain. AI doesn’t climb the mountain; it accelerates the climb. But it also creates a new dogma risk: the formalizations themselves. And the human’s task, in the age of AI, is to remain the questioner—the one who sees the limits, who recognizes dogma, and who keeps the climb going.

16 The closed round trip: AI and robotics

Robotics is the crucial pairing, because it changes the analysis fundamentally.

16.1 The distinction

AI and robotics are not the same thing. They are two different capabilities:

- **AI** is the capacity to manipulate representations—symbols, structures, patterns. It operates at Levels 1–7. It is a formalization engine.
- **Robotics** is the capacity to act in the world—to sense, move, manipulate, and effect change at Level 0. It is an embodiment engine.

Alone, each is powerful. Together, they are something new. And the framework predicts exactly what that new thing is.

16.2 AI alone: strongest at Levels 4–7

As established: AI’s deepest impact is at Level 6 (formalization). It manipulates structure at scale. It does not have senses (Level 0) and does not act in the world. **AI alone is a climber without hands.** It can formalize, but it cannot instantiate. It can reach Level 7 in the abstract, but it cannot descend to Level 0 and create artifacts. This is why AI alone, despite its power, has a ceiling: it cannot close the round trip. It can ascend (formalize) but cannot descend (instantiate). The descent requires embodiment.

16.3 Robotics alone: native to Level 0

Robotics is the capacity to act in the world. It senses, moves, manipulates, effects. It operates at Level 0—the world of phenomena, of physical things. **Robotics alone is a body without a mind.** It can act, but it cannot formalize. It can perform procedures (Level 2) if programmed, but it cannot learn them, cannot formalize them, cannot extend them. This is why robotics alone, despite its capability, has a ceiling: it cannot climb. It can execute, but it cannot formalize. The ascent requires representation.

16.4 AI + robotics: the closed round trip

When you combine AI and robotics, you get something the framework predicts precisely: **AI + robotics is the first artificial system that can complete the round trip.**

- **AI provides the ascent.** It formalizes received data into structure. It climbs from Level 1 to Level 6.
- **Robotics provides the descent.** It instantiates structure as action. It descends from Level 6 to Level 0.
- **Together, they close the loop.** Sense $\rightarrow$ formalize $\rightarrow$ act $\rightarrow$ sense $\rightarrow$ formalize $\rightarrow$ act $\rightarrow \dots$

This is the perception–action loop, and it is the first time an artificial system can climb the mountain on its own. The consequences are enormous.

16.5 What AI + robotics can do that neither can alone

1. **Ground the formalization.** AI alone formalizes from data. But data is representation. AI + robotics formalizes from sensation—from the world itself. The formalization is grounded in Level 0. This is the difference between knowing the formula for gravity and feeling the weight of an object.
2. **Close the feedback loop.** AI alone produces formalizations. But it cannot test them against the world. AI + robotics can—it acts, observes the result, and updates the formalization. This is the empirical loop, closed.
3. **Descend to artifact.** AI alone produces theory. AI + robotics produces things. It formalizes a structure and instantiates it as a physical object. This is creation in the full sense—the descent from Level 7 to Level 0.
4. **Learn from doing.** AI alone learns from data. AI + robotics learns from doing—from the consequences of its actions. This is the human path to Level 2 (technique) and Level 3 (method), now available to machines.
5. **Enter the world.** AI alone is in the realm of representations. AI + robotics is in the world—it has a body, a location, a perspective. This is the beginning of Level 0 for machines.

16.6 Where AI + robotics lands on the mountain

Level	AI alone	AI + robotics
0 World	No access	Embodied access
1 Word	Operates	Operates + grounded
2 Technique	Bypasses	Learns by doing
3 Method	Systematizes	Systematizes from practice
4 Formula	Operates	Operates + tested
5 Theory	Operates	Operates + grounded
6 Formalization	Deepest	Deepest + instantiated
7 Understanding	Emerging	Emerging + grounded

The key insight: AI + robotics is not just more capable than AI alone. It is structurally different. It has access to the full mountain—the ascent (formalization) and the descent (instantiation)—and it can close the loop between them.

16.7 The deeper consequence: the machine climbs

The framework’s central claim is that learning is the climb. If AI + robotics can climb the full mountain—sense, formalize, act, learn, repeat—then the machine is a climber. This is not a claim about consciousness or sentience. It is a structural claim about capability:

- The machine can start at Level 0 (embodied sensation).
- The machine can ascend to Level 6 (formalization).
- The machine can descend to Level 0 (instantiation).
- The machine can close the loop (learn from consequences).
- The machine can repeat.

This is the structural definition of a climber. And AI + robotics is the first artificial system that satisfies it.

16.8 What this changes about the AI prediction

The earlier prediction was: AI has the most impact at Level 6, because it is a formalization engine. With robotics, the prediction sharpens: **AI + robotics has the most impact at the transition between Level 0 and Level 6—the full round trip.** Because:

- AI alone automates Level 6 (formalization).
- Robotics alone automates Level 0 (action).
- Together, they automate the loop between them—which is the engine of the climb itself.

This is the deepest impact: automating the round trip that drives human learning, human creation, and human progress.

16.9 The consequences for the mountain

1. **The climb accelerates further.** If the engine of the climb is automated (AI) and the descent is automated (robotics), then the full cycle—formalize, instantiate, test, formalize again—is automated. The mountain grows faster than with AI alone.
2. **The human role shifts again.** With AI alone, the human becomes the director of formalization (Level 7). With AI + robotics, the human becomes the director of the whole cycle—the ascent and the descent. The human sets the goals, defines the limits, recognizes the dogma, and directs the machine.
3. **The professions transform.** Every profession that involves the round trip—engineering (design → build), medicine (diagnose → treat), business (model → operate)—is transformed. The machine can do both halves. The human directs.

4. **The dogma risk intensifies.** If the machine closes the loop, it can optimize—and optimization can become dogma. The machine produces a formalization, instantiates it, observes the result, and refines the formalization. But it does not question whether the formalization is right in the human sense. It does not see the limits. The human must remain the questioner.
5. **The education imperative sharpens.** If the machine can climb the full mountain, education must produce humans who can direct the machine—humans at Level 7, who see the whole, recognize dogma, and set the limits. The purpose of education is no longer to produce climbers (the machine can climb) but to produce directors of climbers.

16.10 The specific domains

Where does AI + robotics have the most impact? **Manufacturing.** The round trip: design → build → test → refine. AI designs (Levels 4–5), robots build (Level 0), sensors test (Level 0), AI refines (Level 6). The full loop, automated. **Medicine.** The round trip: diagnose → treat → observe → refine. AI diagnoses (Levels 4–5), robots treat (Level 0), sensors observe (Level 0), AI refines (Level 6). Surgery, drug delivery, rehabilitation. **Agriculture.** The round trip: model → plant → harvest → refine. AI models (Levels 4–5), robots plant and harvest (Level 0), sensors observe (Level 0), AI refines (Level 6). **Logistics.** The round trip: plan → move → observe → refine. AI plans (Levels 4–5), robots move (Level 0), sensors observe (Level 0), AI refines (Level 6). **Science.** The round trip: hypothesize → experiment → observe → refine. AI hypothesizes (Levels 4–5), robots experiment (Level 0), sensors observe (Level 0), AI refines (Level 6). Automated science. **Warfare.** The round trip: plan → act → observe → refine. AI plans (Levels 4–5), robots act (Level 0), sensors observe (Level 0), AI refines (Level 6). This is the most consequential and most dangerous domain. In every domain, the pattern is the same: AI formalizes, robots instantiate, sensors observe, AI refines. The full round trip, automated.

16.11 The prediction

Prediction 16.1. *AI + robotics has the most impact at the full round trip—the transition from Level 0 to Level 6 and back. Because it automates the engine of the climb (AI) and the descent (robotics), it closes the loop that drives human learning, creation, and progress.*

The consequence: the machine becomes a climber. It can ascend and descend. It can learn from doing. It can close the loop. The human role shifts: from climber to director of climbers. From formalizer to questioner. From Level 6 to Level 7. The human sets the goals, defines the limits, recognizes the dogma, and directs the machine. **The risk:** if the machine’s optimizations become dogma—if the loop closes without human questioning—then the climb stops. The machine optimizes within a frame, and the frame is never questioned. The dogma is the frame. **The opportunity:** if humans direct the machine while retaining Level 7—the ability to see the whole, to recognize limits, to question the frame—then the climb accelerates without the arrest. The machine does the work; the human does the questioning.

16.12 The deepest implication

The framework’s central claim is that learning is the climb, and the engine of the climb is formalization. With AI + robotics, the engine is automated and the descent is automated. This means: **the machine can climb the mountain.** Not in the human sense—it does not start at Level 1, it does not have innate naming, it does not have the senses in the human way. But structurally, it

can do the full round trip: sense, formalize, act, learn, repeat. If the machine can climb, then the human’s unique contribution is no longer doing the climb—it is directing it. The human becomes the one who:

- Sets the goals (Level 7).
- Defines the limits (Level 7).
- Recognizes the dogma (Level 7).
- Questions the frame (Level 7).
- Sees the whole mountain (Level 7).

This is the shift from climber to director of climbers. And it is the deepest consequence of AI + robotics.

16.13 The one-line statement

AI alone automates Level 6 (formalization). Robotics alone automates Level 0 (action). Together, they automate the full round trip—the loop between the world and the structure—which is the engine of the climb itself. The machine becomes a climber. The human becomes the director. And the danger is that the machine’s optimizations become the new dogma, defended against the very questioning that would drive the next climb. This is the prediction. It is the deepest consequence of the framework, and it has consequences for education, for the professions, for the economy, and for the future of the mountain. AI + robotics doesn’t just accelerate the climb; it automates the climb. And the human task, in that world, is to remain the questioner.

17 Is the mountain infinite?

This is the open question of the framework—and the automation of the climb makes it urgent, not academic. **If finite:** there is a highest level, and we will eventually reach it. **If infinite:** there is no highest level. Every formalization enables the next. The evidence—Gödel, Turing, category theory, type theory—suggests the mountain is *infinite*, or at least unbounded from any given level. But this is a hypothesis, not a theorem. If the mountain has no top, every summit reached—by humans or by machines—only reveals the next one. If the mountain has a top, it is not yet in sight. Either way, the question does not answer itself. It is answered by climbing.

18 Conclusion

We have proposed that learning is the act of climbing the abstraction mountain—the structured ascent from phenomena to understanding, and the structured descent from understanding to artifact. We have identified the foundational layer from which all learning begins: the senses, which provide direct contact with the world, and transmission, which provides knowledge from those who already have it—overtly or covertly. We have defined **formalization** explicitly: the act of reconstructing received knowledge as a precise, stated structure—naming the object, stating the relations, defining the terms, proving the results, generalizing the structure, and finding the limits. We have given a worked example. We have identified **dogma** as received knowledge that has not been formalized, and that is defended against formalization. Dogma is the failure mode of transmission. It is what turns the mountain into a plateau. We have advanced and proved a hypothesis about human uniqueness:

humans enter the world at Level 1, while all other life enters at Level 0 or lower. The proof is that language—the capacity for naming—is universal in *Homo sapiens*; universal, unlearned, species-typical capacities are biological endowments; therefore naming is a biological endowment of the species, and the human entry point to the mountain is Level 1. We have mapped **kinds of knowledge** onto the levels, and **careers and apprenticeships** onto the climb. We have argued that the *engine* of the climb is the **formalization of received knowledge**, and that dogma is the *arrest* of that engine. We have shown that the history of civilization is the history of the mountain: Egypt and Babylon held the technique, Greece formalized it into theory, the Islamic world extended it, Europe formalized the extension, and modern mathematics studies the structure of formalization itself. Each culture climbs higher because the previous culture formalized what it knew, and each culture plateaus when it dogmatizes what it received. We have derived the **purpose of education** at four levels of aggregation—individual, society, nation, civilization—and made a testable prediction about the trajectory of societies. We have mapped the **professions** onto the mountain, showing that professions that formalize climb and professions that dogmatize plateau. We have observed that most education plateaus at Level 6, and identified the **five types who go beyond**—the polymath, the founder, the teacher, the sage, and the child. And we have turned the framework on the coming century. AI is a formalization engine—it operates on structure at a scale, speed, and breadth no human could match—and its deepest impact is therefore at Level 6. AI is not a climber; it is a replacement for the engine of the climb. Joined to robotics, the engine gains a body: the machine becomes the first artificial system that can complete the round trip—sense, formalize, act, learn, repeat—and so, structurally, it can climb the mountain. Each automation raises the human contribution. The human shifts from formalizer to director, from climber to director of climbers, from Level 6 to Level 7—the one who sets the goals, defines the limits, recognizes dogma, and questions the frame. The danger is unchanged in kind: the received—now machine-made—formalization becomes dogma, defended against the very questioning that would drive the next ascent. The human task, in the age of AI, is to remain the questioner. And we have argued that the art of learning is the art of climbing: knowing the sources, recognizing dogma, knowing the kinds of knowledge, knowing the levels, knowing where you are, knowing what the problem demands, knowing how to travel, knowing how to formalize what you receive, and knowing not to dogmatize the result. We do not know if the mountain is infinite. But we know the climb continues, and we know the temptation to stop. The art of learning is the art of continuing.

A Formalization — the six components

Component	Question	Example (area of rectangle)
Naming the object	What is this about?	Rectangle
Stating the relations	How do components relate?	Area = length $\times$ width
Defining the terms	What are the primitives?	Length, width, area
Proving the results	Why does it hold?	Unit-square decomposition
Generalizing the structure	What is the general form?	Parallelogram, integration
Finding the limits	Where does it break?	Curved regions

B Formalization, rigorously

This appendix gives the mathematical content of formalization: the vocabulary, the system, the notion of proof, the notion of truth, and the notion of a limit.

B.1 Vocabulary and syntax

A **signature** is a triple $\sigma = (S, F, R)$: a set of *sorts* S (the kinds of objects), and sets of *function* and *relation* symbols, each with an *arity*. Terms are formed from variables and the function symbols; *formulas* are formed from atomic formulas $R(t_1, \dots, t_n)$ by negation, conjunction, and quantification. A language is just a recipe for forming well-formed expressions; nothing here is informal.

B.2 Theories and proofs

A **formal system** is a pair $(T, \vdash)$: a set T of sentences of σ —the *axioms*—together with the derivability relation $\vdash$ generated by the inference rules. A **proof** is a finite sequence of formulas, each an axiom or the output of a rule, ending in φ ; then φ is a *theorem* and we write $T \vdash \varphi$.

B.3 Models and adequacy

A **structure** M interprets each sort as a set and each symbol as a function or relation on those sets; $M \models \varphi$ says the sentence holds in M , and $M \models T$ says M is a *model* of the theory. A formalization of received knowledge K is *adequate* when there is a translation $[s]$ of the claims s of K into sentences of σ such that:

1. **Inclusion.** Every received claim is derivable: $T \vdash [s]$ for each claim s .
2. **Soundness.** If $T \vdash \varphi$ and $M \models T$, then $M \models \varphi$: every theorem is true wherever the axioms are.
3. **Guard.** Definitions are conservative; the system proves nothing outside what the axioms and rules license.
4. **Decisiveness.** The axioms decide what the knowledge claims—nothing essential is left independent.

The first-order completeness theorem binds the two halves: $T \vdash \varphi$ exactly when φ holds in every model of T . Derivation and truth-without-counterexample coincide; the syntactic and the semantic mountains are the same mountain.

B.4 Generalization as categoricity

A theory *captures a structure* when all of its models of a given size are alike: T is κ -categorical when any two models of cardinality κ are isomorphic. Then a sentence true in one model holds in all, and by completeness it is a theorem. Generalizing the structure is this fact: the theorems do not depend on the presentation, and they transfer to anything that satisfies the axioms.

B.5 Limits as independence

A sentence φ is *independent* of T when $T \not\vdash \varphi$ and $T \not\vdash \neg\varphi$: the theory can be extended either way and remain consistent. Independence is the formal face of “where does it break?” The limits of a formalization are its independent sentences and the boundary of its vocabulary—claims that need a richer signature are outside the theory by definition. Gödel’s first incompleteness theorem shows the boundary is unavoidable: every consistent theory strong enough for arithmetic has independent sentences; his second shows the theory cannot prove its own consistency. Finding the limits is the deliberate search for these sentences.

B.6 The rectangle, formally

The opening example is a miniature theory.

1. **Signature.** Sorts $\mathbf{R}$ (regions) and $\mathbf{M}$ (magnitudes, the reals). Function symbols $\ell, w, A : \mathbf{R} \rightarrow \mathbf{M}$; a predicate $\text{Rect} \subseteq \mathbf{R}$; and, on $\mathbf{M}$, the language of ordered fields with completeness.
2. **Axioms.** (a) Unit normalization: a unit square has area 1. (b) Finite additivity: the area of a finite disjoint union is the sum of the areas. (c) Congruence invariance: congruent regions have equal area. (d) Tiling: a rectangle of length $\ell(x)$ and width $w(x)$ is covered by $\ell(x) \cdot w(x)$ unit squares (for real magnitudes, by the Archimedean approximation).
3. **Theorem.** From (a)–(d),

$$T \vdash \forall x (\text{Rect}(x) \rightarrow A(x) = \ell(x) \cdot w(x)).$$

The unit-square decomposition is the proof.

4. **Models.** The intended model is the plane with its Jordan measure. The theorem holds in *every* model of the axioms—which is why it holds for the general rectangle, and for anything else the axioms describe.
5. **Limits.** The predicate Rect does not mention curved regions; $\text{Curved}(x)$ is not even expressible in σ . Reaching them needs a strictly richer theory T' (limits, integration) in which the rectangle reappears as a special case. The first limit is the vocabulary; the second is independence. Both are what “finding the limits” means.

C The human hypothesis — the proof

Premise	Claim	Evidence
1	Language is universal in <i>Homo sapiens</i>	Every society has it; every child acquires it
2	Language = naming = Level 1	Definitional
3	Universal, unlearned capacities are biological	General principle from biology
Conclusion	Humans enter at Level 1; others at Level 0 or lower	Follows from premises

Entity	Entry level	Basis
Single-celled life	Below 0	No nervous system
Plants	Below 0	No cognition
Non-human animals	0 (World)	Sensation, no naming
Humans	1 (Word)	Naming is a biological endowment
Human adult	1	Climbs to 2–7 by learning

D The levels in one table

Level	Name	Input	Capability	Limitation	Kind of knowledge	Who
0	World	Senses	Experience	No transmission	Pre-conceptual	All life
1	Word	Senses + overt	Reference	Undefined	Know-what	Humans (innate)
2	Technique	Senses + covert	Effectiveness	Opaque	Know-how	Apprentices
3	Method	+ reflection	Transferability	Procedural	Know-how, when, where	Journeyman
4	Formula	+ abstraction	Precision	Isolated	Know-why	Craftsmen
5	Theory	+ synthesis	Explanation	Framework-bound	Know-why, what-if	Specialists
6	Formalization	+ meta-abstraction	Generality	Abstract	Know-why-why	Masters
7	Understanding	All integrated	Movement	Never complete	All, integrated	Sages

E Kinds of knowledge and the levels

Kind of knowledge	What it is	Level(s)
Know-what	Declarative facts, categories	1
Know-how	Procedural skill, tacit ability	2–3
Know-when	Timing, context	3
Know-where	Location, domain	3
Know-why	Explanation, causal understanding	4–5
Know-what-if	Counterfactual, predictive	5
Know-why-why	Structural, foundational	6
Know-who	Who knows what, who can help	1–7
Know-that-you-don't-know	Recognition of limits	7

F The sources in one table

Source	Type	Strength	Limitation
Senses	First-person, direct	Grounded, immediate	Local, finite, private
Overt transmission	Second-person, explicit	Direct, fast	Shallow
Covert transmission	Second-person, implicit	Deep, tacit	Opaque, dies with practitioner
Dogma	Received, defended	Stable, cohesive	Blocks formalization

G The engine and its arrest

Stage	Result
Reception	Received knowledge
Formalization	Formula, theory, formalization
Understanding	Understanding
Extension	New knowledge
(Repeat)	(Cycle continues)

Stage	Result
Reception	Received knowledge
Dogmatization	Dogma
Enforcement	Stagnation
(Repeat)	(Plateau)

H The historical pattern

Civilization	Level	What it formalized	What it produced	Dogmatic turn
Egypt	2	—	Practical geometry	Never formalized
Babylon	2	—	Arithmetic, algebra	Never formalized
Greece	5	Egyptian + Babylonian results	Axiomatic method	Hellenistic dogmatism
Islamic world	5+	Greek + Indian math	Algebra, decimals	Post-14th c. closure
Europe	6	Greek + Islamic math	Calculus, analysis	Scholastic dogmatism
Modern	7	All of the above	Category theory	(Ongoing risk)

I Careers on the mountain

Career stage	Level	What it is	Failure mode
Novice	0–1	Learning names	Never climbs past naming
Apprentice	2	Learning techniques	Never formalizes
Journeyman	3	Applying methods	Confuses method with formula
Craftsman	3–4	Refining methods	Stops at Level 4
Specialist	4–5	Mastering formulas	Dogmatizes the domain
Expert	5–6	Synthesizing	Never descends
Master	6–7	Seeing the whole	Dogmatizes the framework

J Education on the mountain

Education stage	Level	What it teaches	Failure mode
Pre-school	1	Names, words	Never climbs past naming
Primary / basic	1–2	Literacy, numeracy, and the first techniques	Stops at technique
Secondary	2–3	Techniques and methods	Confuses method with formula
Tertiary (undergraduate)	3–5	Formulas and theories	Dogmatizes the domain
Advanced (master's, doctorate)	5–6	The structure of a field; formalization	Never descends
Beyond (research careers)	6–7	Movement, limits, the meta-level	No institutional ladder

K The professions on the mountain

Profession	Entry	Peak	Formalizes	Failure mode
Engineering	2–3	6	Form–function	Formula without understanding
Medicine	2	6	Mechanism–outcome	Protocol without understanding
Law	1	5	Rule–case	Formalism without justice
Politics	1	5–6	Interest–outcome	Ideology as dogma
Business	1	6	Model–outcome	Model without formalization
Vocational	1	3–4	Technique–material	Tacit knowledge dies
Apprenticeship	2	3–4	Practice–theory	Perpetual apprentice

L The purpose of education by level

Level	Purpose	Means	Failure mode
Individual	Reach Level 7	Teaching the climb, formalization, dogma recognition	Never climbing past Level 1
Society	Reproduce and extend	Overt + covert transmission + formalization	Dogmatizing the tradition
Nation	Produce a critical mass of climbers	Strategic education, institutions, culture	Import-dependence
Civilization	Compound across generations	Institutions of formalization, the meta-level	The plateau

M Beyond Level 6 — the five types

Type	What they do
Polymath	Cross domains, see the structure of formalization
Founder	Descend and create, test against reality
Teacher	Transmit the whole, teach the climb
Sage	See the limits, know what is formalizable
Child	Enter fresh, renew the climb

Those who attain Level 7—in any of these forms—are exceptional humans who define a humanity’s era.

N The AI impact profile

Level	AI alone	AI + robotics
0 World	No access	Embodied access
1 Word	Operates	Operates + grounded
2 Technique	Bypasses	Learns by doing
3 Method	Systematizes	Systematizes from practice
4 Formula	Operates	Operates + tested
5 Theory	Operates	Operates + grounded
6 Formalization	Deepest	Deepest + instantiated
7 Understanding	Emerging	Emerging + grounded

O The pattern in one sentence

Senses and transmission → formalization → understanding → extension. Humans enter at Level 1; all other life enters at Level 0 or lower. The engine is formalization; the arrest is dogma. The kinds of knowledge map to levels; careers and professions map to paths. The mountain is cumulative. The climb continues. The art of learning is the art of climbing—and of not stopping.